

Vibhore Mathur

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EDUCATION

B.Tech, Computer Science and Engineering

Swami Keshvanand Institute of Technology, Jaipur

2022 – 2026

CGPA: 8.12/10

Senior Secondary (XII), CBSE

Ryan International School

2022

82.8%

EXPERIENCE

Outbox Labs | Software Developer | Bengaluru, Karnataka

Oct 2025 – Present

- Architected an end-to-end **Model Context Protocol (MCP)** integration with **OAuth 2.0** auth flow and a consent-provider framework enforcing human-in-the-loop validation across **30+ tools**.
- Engineered a centralized **BullMQ + Redis** job queue with retry/backoff for API rate limits, reducing job failures by **40%**.
- Built a high-throughput webhook system processing **15,000+ events/day** with **HMAC verification**, idempotency, and configurable retries.
- Developed a domain health engine for automated **blacklist monitoring** and **DNS validation**, improving email deliverability.
- Spearheaded a **full-scale product revamp** redesigning UI, core workflows, and platform architecture, resulting in improved user engagement and retention.

SoftSensor.AI | MERN Developer Intern | Jaipur, Rajasthan

May 2025 – Aug 2025

- Built a centralized **MQTT alert system** delivering **1,000+ cross-app notifications/week**; collaborated in an **Agile** environment to ship scalable full-stack modules.
- Designed a unified **DTO architecture** across **30+ backend routes**, eliminating inconsistent API responses and reducing client-side defects.

PROJECTS

• CSVQL – Custom SQL Interpreter with Terminal UI



Tools: Python, Textual, Regex, Dataclasses

- Built a **5-stage SQL compiler** (Lexer → Parser → AST → Planner → Executor) from scratch, running queries directly on **CSV files** with zero external dependencies.
- Implemented a **recursive-descent parser** supporting JOINS (INNER/LEFT/RIGHT/FULL), subqueries, GROUP BY, HAVING, and DML/DDL with a **typed AST** of 15+ node types.
- Developed a **Textual TUI** with live schema browser, syntax-highlighted editor, **real-time autocomplete**, and execution-time reporting.

• Stratifyy – Stock Backtesting Platform

[Live Demo](#) |

Tools: React (TypeScript), Node.js, C++, Express.js, PostgreSQL, Socket.io, JWT

- Implemented **Bollinger Bands**, **Mean Reversion**, and **Momentum** strategies with a **C++ computation engine** for high-performance backtesting.
- Built interactive dashboards with equity curve visualization, **real-time trade updates via WebSockets**, and drawdown analysis.

• CineEncode – Distributed Content-Aware Video Transcoder



Tools: Node.js (TypeScript), React, Kafka, BullMQ, Redis, FFmpeg, Socket.io, Groq AI

- Engineered a **distributed transcoding pipeline** that segments video into scene clips using **FFmpeg** change-detection filters, then applies **Groq AI (Llama 3)** to predict optimal CRF values – achieving **40% compression gains** with no perceptual quality loss.
- Built async task processing with **BullMQ/Redis** queues and a **Kafka** event bus, routing real-time encoding progress via **Socket.io WebSockets** to a React dashboard.

SKILLS

- **Languages & Core Technologies**: JavaScript, TypeScript, Python, C++, React.js, Node.js, Express.js, GraphQL, Socket.IO
- **Infrastructure, Databases & Tooling**: PostgreSQL, MongoDB, Redis, AWS, Docker, BullMQ, Qdrant, Browser Automation, Git, OOP